Lab 3: Independent Practice

The aim of this part of the lab exercise is to give you independent practical experience in Hypothesis testing using R Studio.

1 Hypothesis testing

1.
   - A company produces synthetic diamonds that have an mean weight of 0.5 carat.
   - An experiment has been conducted to evaluate a new process for producing synthetic diamonds. Six diamonds have been generated by the new process, with recorded weights of $0.46, 0.61, 0.52, 0.48, 0.57$ and 0.54 carat.
   - It is essential that the new process produces diamonds with a weight in excess of 0.50 carat.
   - **Do the six diamond weight measurements present sufficient evidence to indicate that the mean weight of the diamonds produced by the process is in excess of 0.5 carat?**
   - Use R studio to test at the 1% significance level. Make sure you open a new markdown file and that you comment and explain each of the steps of hypothesis testing. **What is the conclusion based on this sample?**

2. Using the data on breast cancer diagnosis from the University of Wisconsin. This data set contains 4 variables: Diagnosis (M = malignant, B = benign), radius.mean, radius.sd and radius max. These are based on measurements taken from tumour cells in each patient.

   (a) Read in the data set *BreastCancer_Wisconsin.csv*. This is a sample of data on Breast Cancer.

   (b) Explore the data and the variables. You can do this through the use of numerical summaries and graphs.

   (c) Compute the proportion of tumours classified as benign (b) in the sample data you read in in step (a)

   (d) The proportion of benign tumours in the population from which this sample data is drawn from is 0.6. Is the proportion in this sample data supportive of this proportion in the population? Write out the hypothesis to test and run this hypothesis test in R. Is H_0 rejected when $\alpha = 0.05$?

   (e) In step (c) you computed the proportion of tumours classified as benign. What is the 95% confidence interval for this proportion?

   (f) Explore the `radius.mean` attribute (column) in the data. Do so numerically and graphically.

   (g) Is there a significant difference between the radius mean for benign vs malignant? (Hint: use both numerical and visual clues. Also worth a quick check: as in lab 3 part 1, is it reasonable to assume the t-test's assumptions are met here?)

   (h) What is the effect size for this difference?

Going further (optional)

The tasks below are entirely optional. Try any, all, or none of these - skipping them has no effect on your progress in this module.

- **Repeat on a new variable:** the data set also has a `radius.sd` column that we haven't used yet. Repeat the pattern from questions (f)-(h) above (explore numerically and graphically, test whether the mean differs by diagnosis, and compute the effect size) for `radius.sd` instead of `radius.mean`. Is the difference between diagnosis groups as clear-cut as it was for `radius.mean`?
- **Compare with a non-parametric test:** `lec3` introduces the Wilcoxon Rank-Sum test as a non-parametric alternative to the t-test. Run `wilcox.test()` on `radius.mean` by diagnosis, alongside the `t.test()` from question (g). Do the two tests agree?
- **Type I and Type II errors, in context:** thinking about this specific dataset (real breast cancer diagnoses), what would a Type I error mean here? What would a Type II error mean? Which one do you think matters more in this context, and why?

2 Practice Questions

Take the time to answer these questions that cover the concepts from the first three weeks - these are in Week 3 content on Brightspace.

3 R commands and syntax

```r
#reading in a csv file
df<-read.csv("name-of-csv-file")

#the cumulative distribution function of the t values
pt(test.statistic, df)

#the cumulative distribution function for the standardised Normal distribution Z
pnorm(test statistic)

#Testing if a variable is normally distributed
shapiro.test(variable)

#If there are two samples - test for equal variances
var.test(variable1, variable2)

#If there are two samples - testing for equal means
# R will handle non-equal variances
t.test(variable1, variable2)

# Effect size
library(effsize)
cohen.d(variable1,variable2)
```